

SHARAD KUMAR

AI Full Stack Developer | New Delhi, India

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Professional Summary

AI Full Stack Developer with hands-on experience building and deploying **AI-powered, full-stack, and data-driven applications**. Experienced in integrating **LLM-based features, RAG applications, AI agents, intelligent automation, and predictive systems** into web products, with practical experience across frontend, backend, APIs, databases, and cloud deployment. Strong foundation in **software engineering and computer science**, with experience developing production-oriented applications, AI workflows, and end-to-end product solutions.

Technical Skills

Languages: Python, JavaScript/TypeScript, C++, Java
Frontend: React, Next.js, TanStack Start, Vite, Tailwind CSS, Zustand
Backend: Node.js, NestJS, Express.js, FastAPI, REST APIs, JWT Authentication
Databases: PostgreSQL, SQL, SQLite, MongoDB, Redis
AI Frameworks: AI Agents, LLM Applications, RAG, LangChain, scikit-learn, XGBoost
Core Concepts: DSA, OOP, DBMS, Computer Networks, System Design
Other Tools: Docker, Git, GitHub, Postman, Anaconda, Google Colab Vercel, Railway

Work Experience

Airports Authority of India Jun 2026 – Jul 2026
SDE Intern

- Built **Aerodrome**, an ML-powered flight fare prediction system for 6 major routes using demand, seasonality, and date features.
- Engineered the ML pipeline using **Python, XGBoost, and scikit-learn**; served predictions through a **FastAPI** backend.
- Shipped a **React** frontend for fare queries and containerized the application stack using **Docker**.

SSPL, DRDO Sep 2025 – Oct 2025
Software Intern

- Developed a signal-processing pipeline for electromagnetic data using **FFT-based frequency-domain transformation**.
- Improved signal-to-noise ratio by **25–30%** through noise filtering, detrending, and normalization.
- Reduced processing time by **20%** by optimizing computation with **NumPy and Pandas**; analyzed large-scale time-series sensor data.

Projects

ZEKROS – Configurable Strategy Game Platform [Source](#) | [Live](#)
React, Node.js, Express.js, Socket.IO, Prisma, PostgreSQL

- Engineered a multiplayer game platform with a custom game engine for **matches, rooms, players, moves, state, history, and replay**, with persistence via **Prisma/PostgreSQL**.
- Implemented an optimal AI opponent using **Minimax with Alpha-Beta pruning**, with configurable board sizes, game modes, and **Elo-based rating** architecture.
- Built **real-time multiplayer** using **Socket.IO**, synchronizing game state, player actions, and match events across connected clients.

FareLens – AI-Powered Flight Price Prediction & Analytics [Source](#) | [Live](#)
React, Vite, FastAPI, PostgreSQL, SQLAlchemy, XGBoost, SHAP, JWT, Docker

- Engineered a **React SPA** with reusable components, protected routes, REST API integration, and interactive analytics using **Recharts** and Tailwind CSS.
- Integrated **FastAPI REST services and JWT authentication** for prediction workflows, watchlists, price alerts, and authenticated API state handling.
- Integrated **XGBoost inference and SHAP explainability**, visualizing predictions and feature contributions from a model trained on **300K+ flight records**.

CortexFlow – AI Research Assistant [Source](#) | [Live](#)
Python, LangChain, Gemini, Groq, Tavily, BeautifulSoup, Streamlit

- Engineered a **4-stage AI research pipeline** using LangChain to orchestrate Search, Read, Write, and Critique stages for automated web research and report generation.
- Integrated **Tavily web search** and **BeautifulSoup scraping** to retrieve and preprocess source content before LLM-based synthesis.
- Designed **multi-provider LLM fallback** with Gemini/Groq and automated critique, producing structured reports with **PDF export**.

Certificate

Oracle Agentic AI Foundations (2026) [\[View Certificate\]](#)

Education

Guru Gobind Singh Indraprastha University – GTB4CEC, Delhi Aug 2023 – Jul 2027
B.Tech in Computer Science and Engineering **CGPA: 8.23**